

Supplementary for An Efficient Insect-inspired Approach for Visual Point-goal Navigation

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I. MODEL DETAILS

More details of the user-defined parameters in our insect-inspired model are summarised in Table S1. We did not attempt to optimise the model: α was set to be the highest possible learning rate without any tuning, and most other parameters were chosen to be the same as in our earlier work [1], where they were hand-tuned but for the different model. Note τ_d and τ_c were set to different values in the two simulators to adapt to the different low-level motion control mechanisms (§II-B3).

TABLE S1
MORE DETAILS OF THE INSECT-INSPIRED MODEL.

variable	meaning	Habitat	iGibson	remarks
N_{PN}	number of PNs	1089	1089	determined by preprocessed image size [1]
$N_{PN/KC}$	average number of PNs connected to every KC	10	10	approximating real PN-KC connectivity [1]
N_{KC}	number of KCs	32000	32000	optimised in [2] for our earlier work [1]
k	number of activated KCs	320	320	k/N_{KC} approximating real KC sparsity [1]
α	learning rate of KC-MBON weights	1	1	highest possible rate
τ_d	epochs between visual and collision-related inputs	1	6	minimal hand-tuning
τ_c	epochs of escape	$2 \left\lfloor \frac{ \gamma_c - 90^\circ }{10^\circ} \right\rfloor$	15	minimal hand-tuning
F_c	force threshold for collision	—	100 N	minimal hand-tuning [1]

More details of the seven recent point-goal navigation models cited in Table II are summarised in Table S2. As [3] used a different simulator with a different dataset of scene reconstructions, the performance in Fig. 2 is inflated by a factor of about 1.3 from the value in this table. The factor is calculated as the ratio of the different results of the same model in Gibson and MP3D scenes reported in Table 2 of [4].

TABLE S2
MORE DETAILS OF THE SEVEN MODELS IN TABLE II.

Model	Camera	Vision Model	RL /SLAM Model	Training Scenes	Training Epochs	SR	SPL	Reference
S1	RGB	CNN	ORB-SLAM2, D* Lite	SunCG, MP3D	5×10^6	0.090	0.047	Table 3 & 11 [3]
S2	RGBD	CNN	inherited from S1	Gibson 4+, MP3D	7.5×10^7	0.62	0.51	Table 2 [4]
V1	noisy RGB	ResNet18	inherited from E2	Gibson 4+	1.47×10^7	0.59	0.45	Table 2 [5]
V2	noisy RGBD	ResNet50	inherited from E2	Gibson 0+	7.5×10^9	0.960	0.766	Table 1 [6]
E1	RGB	same as S2	PPO	same as S2	7.5×10^7	0.64	0.46	Table 2 [4]
E2	RGB	SE-ResNeXt50	DDPPO	Gibson 2+, MP3D	2.5×10^9	0.977	0.920	Table 2 [7]
E3	RGB	ResNet50	inherited from E2	Gibson 0+	1.5×10^9	0.96	0.87	Table 2 [8]

II. SUPPLEMENTARY FIGURES

See Fig. S1-S4.

REFERENCES

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- [8] S. K. Ramakrishnan, A. Gokaslan, E. Wijmans, O. Maksymets, A. Clegg, J. Turner, E. Undersander, W. Galuba, A. Westbury, A. X. Chang *et al.*, “Habitat-Matterport 3D dataset (HM3D): 1000 large-scale 3D environments for embodied AI,” *arXiv preprint arXiv:2109.08238*, 2021.

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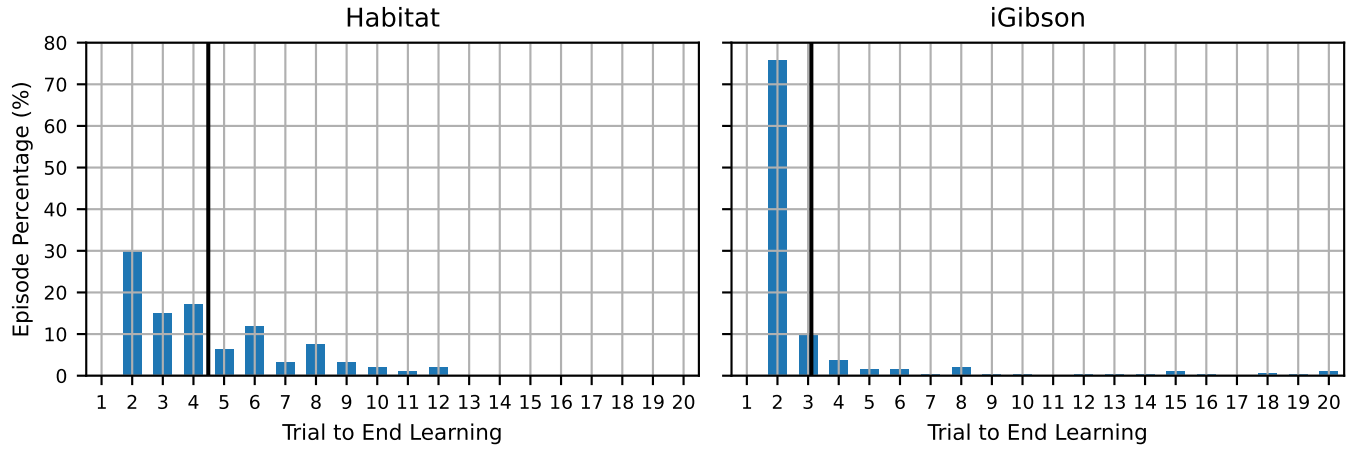


Fig. S1. Trial when the insect-inspired model decided to end learning as drop of SPL was detected and became 'Learnt'. The vertical lines denote the average trial index, approximately 4.5 and 3.1 in Habitat and iGibson, respectively.

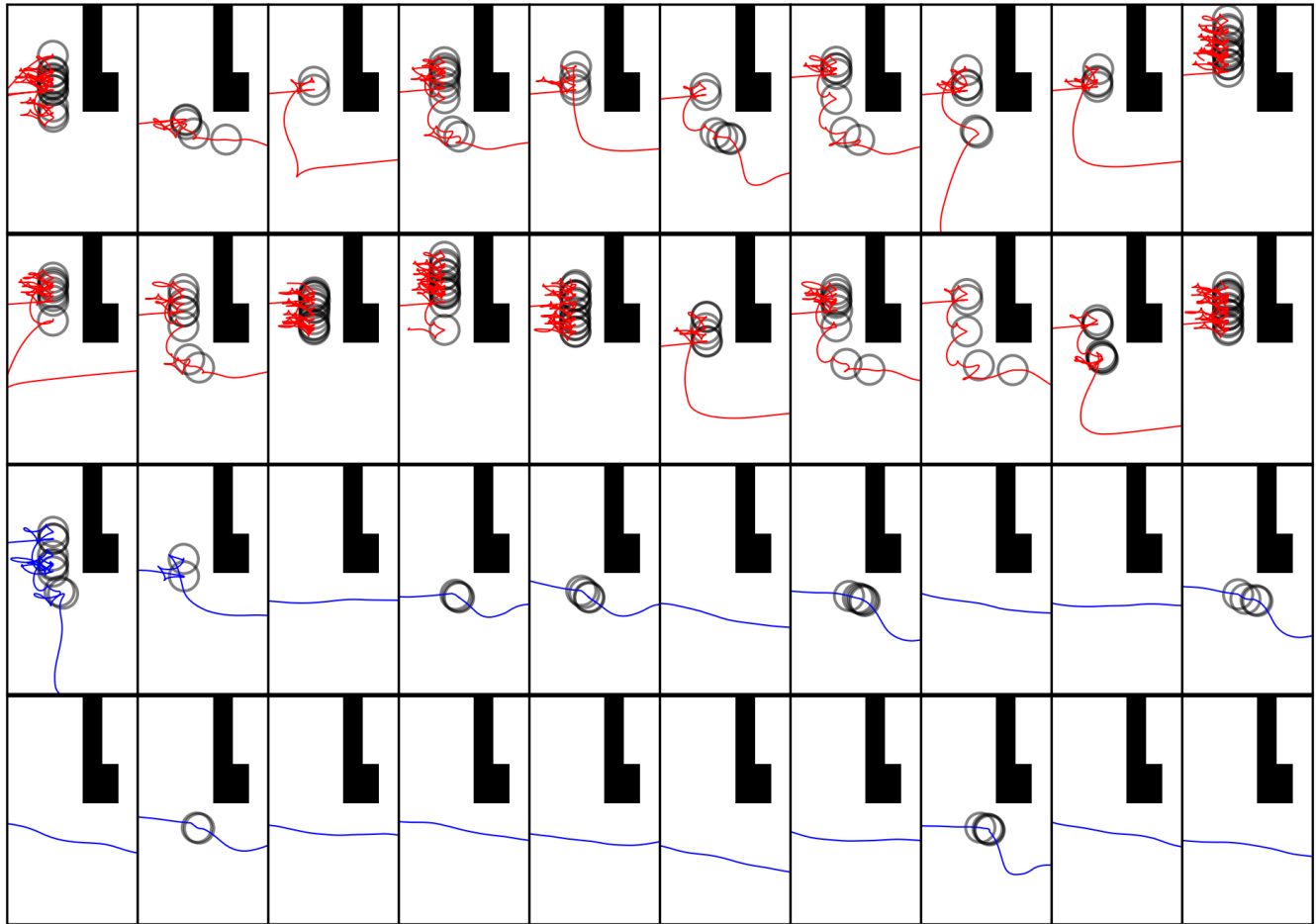


Fig. S2. Zoom-in plots of the area enclosed in the rectangle with dashed boundaries in Fig. 4. Blue: the insect-inspired model (full). Red: the odometry-collision model (A2). The black circles here denote where collisions were detected.

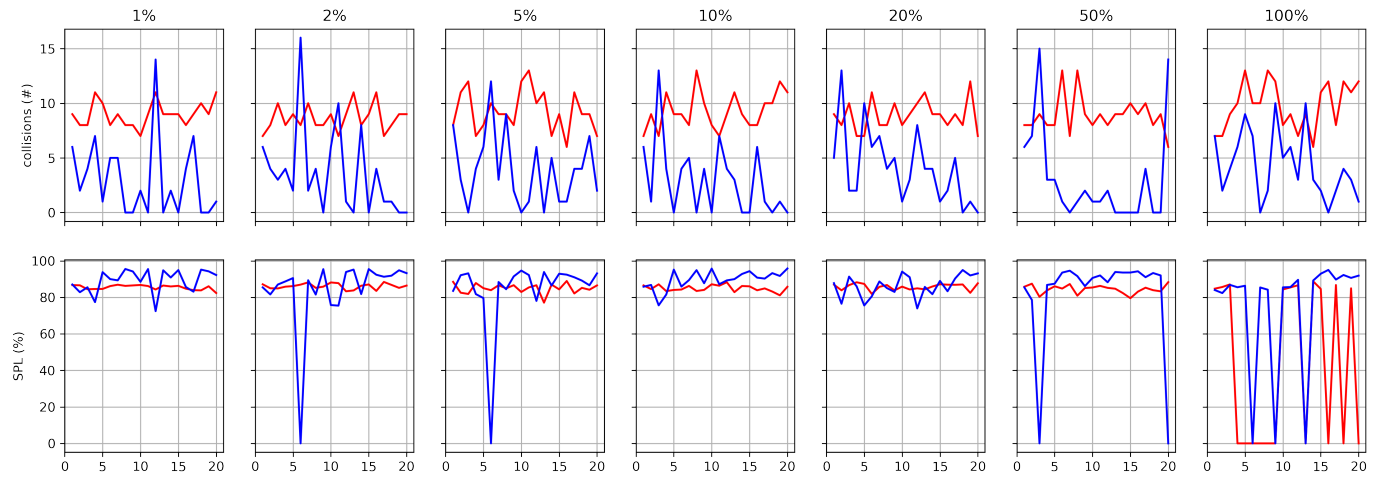


Fig. S3. The effect of motor noise on performance, across trials within an episode. Blue: the insect-inspired model (full). Red: the odometry-collision model (A2).

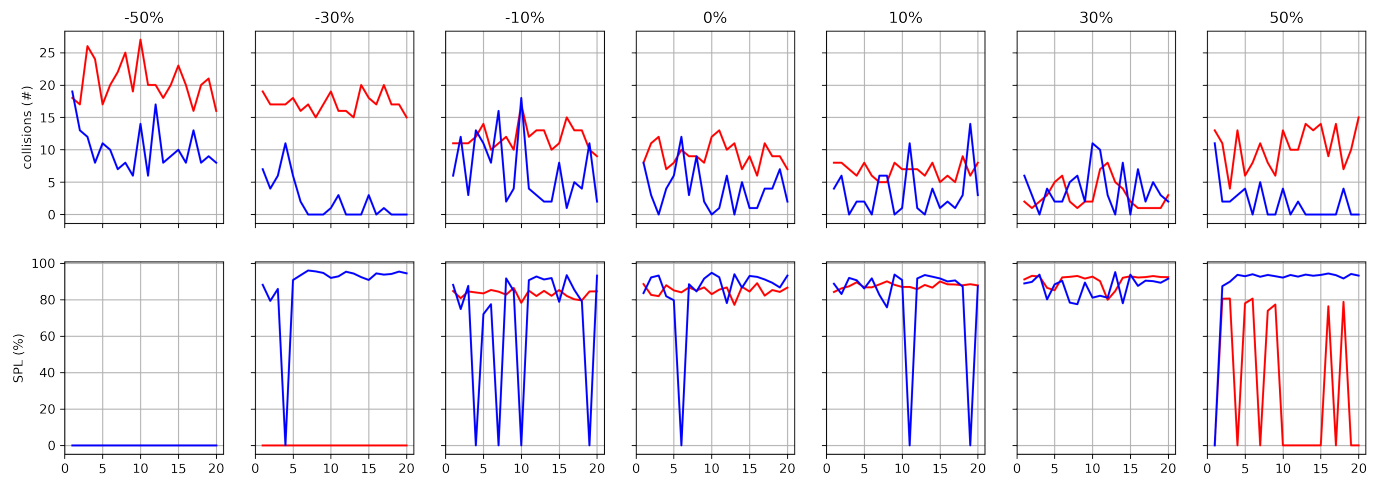


Fig. S4. The effect of motor bias on performance, across trials within an episode. Blue: the insect-inspired model (full). Red: the odometry-collision model (A2).